DoD 8140 Cybersecurity Workforce Qualification — Personal Certification Path

[PLACEHOLDER — narrative to be rewritten by Jeff]

This workbook presents the certification-path options from DoDM 8140.03, sourced from the DoD 8140 Foundational Qualification Matrix V2.1 (effective 2025-09-19). Other qualification paths in the DoD 8140 paradigm (education, DoD training, commercial training, experience) are intentionally out of scope for this reference.

Tabs:
- 'Certification Requirements' — per-role summary: for each work role, the approved certification options at Basic / Intermediate / Advanced levels.
- 'Certification Analysis' — inverted pivot view: all certs across all work roles on one page. Cells contain the highest proficiency level (1 = Basic, 2 = Intermediate, 3 = Advanced) at which that cert qualifies a person for that role. Summary rows at the bottom.

Work roles with no certification data (pending DoD review):
- (462) Control Systems Security Specialist
- (731) Cyber Legal Advisor
- (901) Executive Cyber Leader

These roles exist in the DoD 8140 V2.1 role universe but have no published certification options. Omitted from the matrix; will be added when DoD publishes data.

Authoritative source: DoD 8140 Foundational Qualification Matrix V2.1
URL at time of refresh: www.cyber.mil/dod-workforce-innovation-directorate/dod8140/qualification-matrices

Compiled by Jeff Krueger.

Certification Path Qualification Options (DoD 8140.03)

Work Role	Basic	Intermediate	Advanced
(111) All-Source Analyst			CySA+, GCTI, RCCE Level 1
(121) Exploitation Analyst			GCIH, GPEN, PenTest+
(131) Joint Targeting Analyst			GPEN
(132) Target Digital Network Analyst			GCIH, GPEN
(211) Forensics Analyst		CHFI, PenTest+	CFR, CySA+, GCFA, GCFE, RCCE Level 1
(212) Cyber Defense Forensics Analyst		CHFI, RCCE Level 1	CFR, CySA+, GCFA, GCFE, GREM, PenTest+
(221) Cyber Crime Investigator	GCIH	CHFI, PenTest+	CFR, CySA+, GCFA, GCFE, SecurityX / CASP+
(311) All-Source Collection Manager			GCFA, GCTI
(312) All-Source Collection Requirements Manager			GCFA, GCTI
(331) Cyber Intelligence Planner			GCTI
(332) Cyber Operations Planner			GCIH, GPEN, RCCE Level 1
(411) Technical Support Specialist	A+, Network+	CND, GFACT, GSEC, Security+	CCNP Security, CISA, FITSP-O, GICSP, SSCP, SecurityX / CASP+
(421) Database Administrator		Cloud+, GSEC, SSCP, Security+	CCNP Security, CISA, CISSP, CISSP-ISSAP, CISSP-ISSEP, SecurityX / CASP+
(422) Data Analyst		GSEC, SSCP	CCISO, CISM, CISSP, GSLC
(431) Knowledge Manager		SSCP, Security+	
(441) Network Operations Specialist	CND, Network+	CEH, Cloud+, GCIH, GICSP, GSEC, SSCP, Security+	CCNA, CCNP Security, CCSP, GCED, GCIA, GCLD, GDSA, GFACT, SecurityX / CASP+
(451) System Administrator	A+, CND, Network+	Cloud+, GICSP, GSEC, SSCP, Security+	CCNP Security, CCSP, FITSP-O, GFACT, SecurityX / CASP+
(461) Systems Security Analyst	CND, SSCP	CCSP, Cloud+, GICSP, GISF, GSEC, Security+	CISSO, CISSP-ISSEP, CySA+, FITSP-O, GCLD, GCSA, GSNA, RCCE Level 1
(511) Cyber Defense Analyst	CC, CEH, GFACT, GISF	CEH(P), Cloud+, FITSP-O, GCED, GDSA, GMON, GRID, GSEC, PenTest+, Security+	CBROPS, CFR, CySA+, GCFA, GCIA, GICSP
(521) Cyber Defense Infrastructure Support Specialist	A+, CC, CND, GCLD, GDSA, GFACT, Network+	CEH, Cloud+, CySA+, GMON, GRID, GSEC, PenTest+, SSCP, Security+	CISSP-ISSAP, CISSP-ISSEP, GCIA, GICSP
(531) Cyber Defense Incident Responder	CC, GDSA, GISF	CBROPS, CCSP, CEH, CEH(P), Cloud+, ECIH, FITSP-O, GCED, GCIH, GRID, GSEC, PenTest+, RCCE Level 1, Security+	CFR, CySA+, GCFA, GCIA, GICSP
(541) Vulnerability Assessment Analyst	CEH	CEH(P), CPTE, Cloud+, FITSP-A, GCED, GCIH, GCSA, GICSP, GSEC, PenTest+, RCCE Level 1, Security+	CFR, CISA, CISM, CySA+, GPEN, GSNA
(611) Authorizing Official/Designated Representative		CCSP, CGRC/CAP, Cloud+, GSEC	CCISO, CISM, CISSP, CISSP-ISSEP, CISSP-ISSMP, FITSP-M, GCSA, GSLC
(612) Security Control Assessor		CGRC/CAP, CISSO, Cloud+, FITSP-A, GCSA, GSEC, PenTest+, Security+, SecurityX / CASP+	CCISO, CISA, CISM, CISSP, CISSP-ISSEP, CySA+, GSLC, GSNA
(621) Software Developer		CSSLP, GSEC	CISSP-ISSAP
(622) Secure Software Assessor	GCLD	CSC, CSSLP, GCSA, GSEC, Security+	CISSP-ISSEP
(631) Information Systems Security Developer	CC, CND, GISF, SSCP	CCSP, CSC, Cloud+, GCLD, GCSA, GSEC, SecurityX / CASP+	CISSP-ISSEP, FITSP-D
(632) Systems Developer	CND	CSSLP, GCLD, GSEC	FITSP-D, GCSA, GISF, SSCP
(641) Systems Requirements Planner	CND	CCSP, CFR, FITSP-M, GSEC, SSCP, Security+	GCSA, GSLC, SecurityX / CASP+
(651) Enterprise Architect	CND	CCSP, CSSLP, Cloud+, FITSP-D, GDSA, GSEC, SecurityX / CASP+	CISSO, CISSP-ISSAP, CISSP-ISSEP, GCIA, GCLD, GCSA, GICSP
(652) Security Architect	GCLD, GISF	CCSP, CISSO, CSSLP, Cloud+, FITSP-D, GCSA, GMON, GSEC, SecurityX / CASP+	CCNP Enterprise, CISM, CISSP-ISSAP, CISSP-ISSEP, GCIA, GDSA, GICSP
(661) Research & Development Specialist		Security+	CEH, GCLD, RCCE Level 1, SecurityX / CASP+

(671) System Testing and Evaluation Specialist	CND, Network+	CEH, CFR, Cloud+, GSEC, PenTest+, SSCP, Security+	CCSP
(711) Cyber Instructional Curriculum Developer		GSEC	
(712) Cyber Instructor			CISSO
(722) Information Systems Security Manager	CC	CCISO, CCSP, CGRC/CAP, CISSO, Cloud+, GCSA, GMON, GSEC, SSCP, Security+, SecurityX / CASP+	CISM, CISSP, CISSP-ISSMP, FITSP-M, GCIA, GCIH, GICSP, GSLC
(723) COMSEC Manager		GSEC	CISM, CISSO, FITSP-M, GCIH, GCSA, GICSP, GSLC
(732) Privacy Compliance Manager			CISSO, GSLC
(751) Cyber Workforce Developer and Manager		Security+	CCISO, CCSP, CFR, CISM, CISSP, GSLC, SecurityX / CASP+
(752) Cyber Policy and Strategy Planner		Security+	CCISO, CCSP, CISM, CISSO, CISSP, GSLC, SecurityX / CASP+
(801) Program Manager		CGRC/CAP, DAWIA PM Practioner, SecurityX / CASP+	CCISO, CISM, CISSO, CISSP, DAWIA PM Advanced, GFACT, GSLC, RCCE Level 1
(802) IT Project Manager		DAWIA PM Practioner, GSEC, Security+, SecurityX / CASP+	CCISO, CCSP, CISA, CISM, CISSP, CISSP-ISSEP, CISSP-ISSMP, DAWIA PM Advanced, FITSP-A, FITSP-M, GFACT, GSLC, RCCE Level 1
(803) Product Support Manager		DAWIA LCL Foundational	CCISO, DAWIA LCL Advanced, GCSA, GFACT, GISF, GSLC, RCCE Level 1
(804) IT Investment/Portfolio Manager			CCISO, CISM, CISSO, CISSP, FITSP-A, FITSP-M, GCSA, GFACT, GSLC
(805) IT Program Auditor		CGRC/CAP, GSEC, GSNA, SSCP, Security+, SecurityX / CASP+	CCISO, CCSP, CISA, CISM, CISSP, FITSP-A, FITSP-M, GCSA, GISF, GSLC, RCCE Level 1

Note: The following work roles exist in DoD 8140 V2.1 but have no published cert options (pending DoD review): (462) Control Systems Security Specialist, (731) Cyber Legal Advisor, (901) Executive Cyber Leader.

DoD 8140.03 Foundational Qualification: Personal Certification

Work Role	CompTIA							RCCE	EC-Council						FITSI				GIAC (SANS)														ISACA		(ISC)2										CertNexus		CISCO				mile2		DAWIA				Work Role
	A+	Net+	Cloud+	Sec+	PenTest	SecX	CySA+	RCCE-1	CND	CEH	CEH(P)	CHFI	ECIH	CCISO	FITSP-D	FITSP-A	FITSP-O	FITSP-M	GISF	GDSA	GMON	GRID	GSEC	GCLD	GCED	GCIH	GFACT	GCSA	GICSP	GSNA	GCFA	GCFE	GCIA	GCTI	GPEN	GREM	GSLC	CISA	CISM	CC	CGRC	CSSLP	SSCP	CCSP	CISSP	ISSAP	ISSEP	ISSMP	CSC	CFR	CBROPS	CCNA	CCNP-E	CCNP-S	CPTE	CISSO	LCL-F	PM-P	LCL-A	PM-A	
(111) All-Source Analyst							3	3																										3																										(111) All-Source Analyst
(121) Exploitation Analyst					3																					3									3																									(121) Exploitation Analyst
(131) Joint Targeting Analyst																																			3																									(131) Joint Targeting Analyst
(132) Target Digital Network Analyst																										3									3																									(132) Target Digital Network Analyst
(211) Forensics Analyst					2		3	3				2																		3	3																			3											(211) Forensics Analyst
(212) Cyber Defense Forensics Analyst					3		3	2				2																		3	3					3														3											(212) Cyber Defense Forensics Analyst
(221) Cyber Crime Investigator					2	3	3					2														1				3	3																			3											(221) Cyber Crime Investigator
(311) All-Source Collection Manager																														3				3																										(311) All-Source Collection Manager
(312) All-Source Collection Requirements Manager																														3				3																										(312) All-Source Collection Requirements Manager
(331) Cyber Intelligence Planner																																		3																										(331) Cyber Intelligence Planner
(332) Cyber Operations Planner								3																		3									3																										(332) Cyber Operations Planner
(411) Technical Support Specialist	1	1		2		3			2								3						2				2		3								3					3												3						(411) Technical Support Specialist
(421) Database Administrator			2	2		3																	2														3					2		3	3	3								3						(421) Database Administrator
(422) Data Analyst														3									2													3		3					2		3															(422) Data Analyst
(431) Knowledge Manager				2																																						2																		(431) Knowledge Manager
(441) Network Operations Specialist		1	2	2		3			1	2										3			2	3	3	2	3		2				3										2	3							3			3						(441) Network Operations Specialist
(451) System Admin (req'd for admin access)	1	1	2	2		3			1								3						2				3		2														2	3										3						(451) System Admin (req'd for admin access)
(461) Systems Security Analyst			2	2			3	3	1								3		2				2	3				3	2	3													1	2			3										3			(461) Systems Security Analyst
(511) Cyber Defense Analyst			2	2	2		3			1	2						2		1	2	2	2	2		2		1		3		3		3						1											3	3									(511) Cyber Defense Analyst
(521) Cyber Defense Infrastructure Support Specialist	1	1	2	2	2		2		1	2										1	2	2	2	1			1		3				3						1				2			3	3														(521) Cyber Defense Infrastructure Support Specialist
(531) Cyber Defense Incident Responder			2	2	2		3	2		2	2		2				2		1	1		2	2		2	2			3		3		3						1					2						3	2									(531) Cyber Defense Incident Responder
(541) Vulnerability Assessment Analyst			2	2	2		3	2		1	2					2							2		2	2		2	2	3					3			3	3											3						2						(541) Vulnerability Assessment Analyst
(611) Authorizing Official/Designated Representative			2											3				3					2					3								3		3		2			2	3		3	3														(611) Authorizing Official/Designated Representative
(612) Security Control Assessor			2	2	2	2	3							3		2							2					2		3						3	3	3		2			3		3											2					(612) Security Control Assessor
(621) Software Developer																							2																		2				3															(621) Software Developer
(622) Secure Software Assessor				2																			2	1				2													2					3		2												(622) Secure Software Assessor
(631) Information Systems Security Developer			2			2			1						3				1				2	2				2											1			1	2				3		2												(631) Information Systems Security Developer
(632) Systems Developer									1						3				3				2	2				3													2	3																			(632) Systems Developer
(641) Systems Requirements Planner				2		3			1									2					2					3								3						2	2						2											(641) Systems Requirements Planner
(651) Enterprise Architect			2			2			1						2					2			2	3				3	3				3									2		2		3	3									3				(651) Enterprise Architect
(652) Security Architect			2			2									2				1	3	2		2	1				2	3				3					3				2		2		3	3					3				2				(652) Security Architect
(661) Research & Development Specialist				2		3		3			3													3																																			(661) Research & Development Specialist
(671) System Testing and Evaluation Specialist		1	2	2	2				1	2													2																			2	3						2											(671) System Testing and Evaluation Specialist
(711) Cyber Instructional Curriculum Developer																							2																																					(711) Cyber Instructional Curriculum Developer
(712) Cyber Instructor																																																								3				(712) Cyber Instructor
(722) Information Systems Security Manager			2	2		2								2				3			2		2			3		2	3				3				3		3	1	2		2	2	3			3								2					(722) Information Systems Security Manager
(723) COMSEC Manager																		3					2			3		3	3							3		3																		3					(723) COMSEC Manager
(732) Privacy Compliance Manager																																					3																		3					(732) Privacy Compliance Manager
(751) Cyber Workforce Developer and Manager				2		3								3																							3		3				3	3						3										(751) Cyber Workforce Developer and Manager
(752) Cyber Policy and Strategy Planner				2		3								3																							3		3				3	3												3					(752) Cyber Policy and Strategy Planner
(801) Program Manager						2		3						3													3										3		3			2			3											3		2		3	(801) Program Manager
(802) IT Project Manager				2		2		3						3		3		3					2				3										3	3	3					3	3			3	3								2		3	(802) IT Project Manager
(803) Product Support Manager								3						3					3								3	3									3																					2		3		(803) Product Support Manager
(804) IT Investment/Portfolio Manager														3		3		3									3	3									3		3							3										3					(804) IT Investment/Portfolio Manager
(805) IT Program Auditor				2		2		3						3		3		3	3				2					3		2							3	3	3		2		2	3	3																	(805) IT Program Auditor
Proficiency levels: Basic = 1, Intermediate = 2, Advanced = 3	A+	Net+	Cloud+	Sec+	PenTest	SecX	CySA+	RCCE-1	CND	CEH	CEH(P)	CHFI	ECIH	CCISO	FITSP-D	FITSP-A	FITSP-O	FITSP-M	GISF	GDSA	GMON	GRID	GSEC	GCLD	GCED	GCIH	GFACT	GCSA	GICSP	GSNA	GCFA	GCFE	GCIA	GCTI	GPEN	GREM	GSLC	CISA	CISM	CC	CGRC	CSSLP	SSCP	CCSP	CISSP	ISSAP	ISSEP	ISSMP	CSC	CFR	CBROPS	CCNA	CCNP-E	CCNP-S	CPTE	CISSO	LCL-F	PM-P	LCL-A	PM-A	*darker shades cover more work roles*
Total Positions Covered	3	5	15	**20**	10	17	10	12	10	7	3	3	1	11	4	5	5	7	8	6	4	3	**25**	9	4	9	9	15	12	4	7	3	7	4	5	1	14	6	13	5	5	5	14	15	11	5	10	3	2	9	2	1	1	4	1	11	1	2	1	2	**Total Positions Covered**
Total "Points" (proficiency levels × positions)	3	5	30	**40**	22	**43**	29	33	11	13	6	6	2	32	10	13	13	20	15	12	8	6	**50**	19	9	22	22	**39**	32	11	21	9	21	12	15	3	**42**	18	**39**	5	10	10	28	**37**	33	15	30	9	4	25	5	3	3	12	2	30	2	4	3	6	**Total "Points"**

Note: The following work roles exist in DoD 8140 V2.1 but have no published certification options (pending DoD review): (462) Control Systems Security Specialist, (731) Cyber Legal Advisor, (901) Executive Cyber Leader.